

Math

Mashud Sir

Laplace transform of a function division by t .
 If

$$L\{F(t)\} = f(s) \quad \text{Then} \quad L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(u) du$$

Laplace transform of integrals

$$(i) \quad L\left\{\int_0^\infty F(u) du\right\} = \frac{f(s)}{s}$$

$$(ii) \quad L\left\{\int_0^t \frac{\sin u}{u} du\right\} = \frac{1}{s} \tan^{-1} \frac{1}{s}$$

[Derivim nika
 na - sir]

Inverse Laplace transform

If $L\{F(t)\} = f(s)$ then $F(t)$ is called an inverse Laplace transform of $f(s)$ and it is denoted by $F(t) = L^{-1}\{f(s)\}$ $L^{-1} \rightarrow$ operator.

Ex:

$$(i) \quad L^{-1}\left\{\frac{1}{s-1}\right\} = e^t$$

$$(ii) \quad L^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos at$$

$$(iii) \quad L^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$(iv) \quad L^{-1}\left\{\frac{2!}{s^3}\right\} = t^2$$

[जानक result
 भुझ कर
 शक]

Question

- * Mashud sir ka part repeat kar na
- * exercise book ka exercise kar na kar na i solution solve kar na kar na



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Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date :/...../.....

$$* \mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} = \frac{1}{2!} t^2$$

$$\text{EX: } \mathcal{L}^{-1} \left\{ \frac{s}{s-3} - \frac{2s}{s^2+25} + \frac{3}{s^2+9} + \frac{1}{s^4} \right\}$$

$$= 5e^{3t} - 2\cos 5t + \frac{3}{2} \sin 2t + \frac{1}{6} t^3$$

$$\mathcal{L}^{-1} \left\{ \frac{5}{s-3} \right\}$$

$$= 5 \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\}$$

$$= 5e^{3t}$$

$$\mathcal{L}^{-1} \left\{ \frac{2s}{s^2+25} \right\}$$

$$= 2 \mathcal{L}^{-1} \left\{ \frac{s}{s^2+5^2} \right\}$$

$$= 2\cos 5t$$

$$\mathcal{L}^{-1} \left\{ \frac{3}{s^2+9} \right\}$$

$$= \frac{3}{2} \mathcal{L}^{-1} \left\{ \frac{2}{s^2+3^2} \right\}$$

$$= \frac{3}{2} \sin 2t$$

268 page

$$\textcircled{1} \mathcal{L}^{-1} \left\{ \frac{6s-4}{s^2-4s+20} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{6s-4}{(s-2)^2+4^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{6(s-2)+8}{(s-2)^2+4^2} \right\}$$

$$= 6 \mathcal{L}^{-1} \left\{ \frac{s-2}{(s-2)^2+4^2} \right\} + \mathcal{L}^{-1} \left\{ \frac{2 \cdot 4}{(s-2)^2+4^2} \right\}$$

$$= 6 \mathcal{L}^{-1} \left\{ \frac{s-2}{(s-2)^2+4^2} \right\} + 2 \mathcal{L}^{-1} \left\{ \frac{4}{(s-2)^2+4^2} \right\}$$

$$= 6 \cdot e^{2t} \cos 4t + 2 \cdot e^{2t} \sin 4t$$

$$= 2 \cdot e^{2t} (3 \cos 4t + \sin 4t)$$

अ रूपांतर 1 की शक है।
partial शक है

003

Plan Title: 1-12-20

Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date: / /

lecture - 06

Math

Mashud Sir

marks
test sheet

$$y''(x) - y'(x) = x \text{ when } y(0) = 2$$

Soln:

$$y''(x) - y'(x) = x$$

$$\hookrightarrow \{ y''(x) - y'(x) \} = \{ x \}$$

$$s^2 Y(s) - sY(0) - y'(0) = \frac{1}{s^2}$$

$$Y(s) \{ s^2 - s \} = \frac{1}{s^2} + 2s - 5$$

$$Y(s) = \frac{1}{s^2(s^2-s)} + \frac{2s}{s^2-s} - \frac{5}{s^2-s}$$

$$Y(s) = \frac{1}{s^2(s^2-s)} + \frac{2s}{s^2-s} - \frac{5}{s^2-s}$$

$$= \frac{1}{s^3(s-1)} + \frac{2}{s-1} - \frac{5}{s(s-1)}$$

$$= -\frac{1}{s} - \frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{s-1} + \frac{2}{s-1} + \frac{5}{s} - \frac{5}{s-1}$$

partial
fraction
2/4

$$= \frac{4}{s} - \frac{1}{s^2} - \frac{1}{s^3} - \frac{2}{s-1}$$

$$\mathcal{L}^{-1} \{ Y(s) \} = \mathcal{L}^{-1} \left\{ \frac{4}{s} - \frac{1}{s^2} - \frac{1}{s^3} - \frac{2}{s-1} \right\}$$

$$Y(x) = 4 \cdot 1 - 1 \cdot x - \frac{1}{2!} x^2 - 2e^x$$

$$= 4 - x - \frac{x^2}{2} - 2e^x$$

V.V.I Ex:

$$Y''(t) + 2Y'(t) = \cos 2t \quad Y(0)=1, \quad Y\left(\frac{\pi}{2}\right)=-1$$

Sol:

$$\mathcal{L} \{ F(t) \} = f(s)$$

$$\mathcal{L} \{ F'(t) \} =$$

$$\mathcal{L} \{ F''(t) \} = ?$$

$$\mathcal{L}^{-1} \{ f(s) \} = F(t)$$

$$\mathcal{L} \{ y(t) \} = Y(s)$$

$$y'(0) = a$$

$$Y(s) = \frac{1}{s} + \frac{s}{s^2+4} + \frac{9}{s} + \frac{s}{s^2+9} + \frac{9}{s^2+9}$$

$$\mathcal{L}^{-1} \{ Y(s) \} =$$

$$y(t) =$$



Plan Title :

Sat / Sun / Mon / Tue / Wed / Thu / Fri
☐ ☐ ☐ ☐ ☐ ☐ ☐

Date :/...../.....

$$y(t) = \frac{1}{5} \cos 2t + \frac{4}{5} \cos 3t + \frac{9}{3} \sin 3t$$

$$y(0) = 1 \quad y\left(\frac{\pi}{2}\right) = -1$$

$$y\left(\frac{\pi}{2}\right) = \frac{1}{5} \cos 2 \cdot \frac{\pi}{2} + \frac{4}{5} \cos 3 \cdot \frac{\pi}{2} + \frac{9}{3} \sin 3 \cdot \frac{\pi}{2}$$

$$-1 = -\frac{1}{5} + 0 + \frac{9}{3} (-1)$$

~~something~~

$$a = \frac{12}{5}$$

then Final Result
 Laplace + Inverse Laplace [xcs]

2nd order derivative

1st order math & solve

normal calculation

integration + derivative = solve math

Math

Mashud Sir

03-12-20

ত্রিকোণমিতিক ধারা

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$\therefore a_0, a_n$ and b_n constant

* (i) $\int_{-\pi}^{\pi} \sin nx dx = 0$

(ii) $\int_{-\pi}^{\pi} \cos nx dx = 0$

(iii) $\int_{-\pi}^{\pi} \sin nx \cos nx dx = 0$

(iv) $\int_{-\pi}^{\pi} \sin nx \sin mx dx = 0$

(v) $\int_{-\pi}^{\pi} \cos nx \cos mx dx = 0$

(vi) $\int_{-\pi}^{\pi} \cos^2 nx dx = \pi$

(vii) $\int_{-\pi}^{\pi} \sin^2 nx dx = \pi$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

show that,

$$(1) a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$(2) a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$(3) b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Soln:

both side
integration
limit $-\pi \rightarrow \pi$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$\int_{-\pi}^{\pi} f(x) dx = \frac{a_0}{2} \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos nx dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin nx dx$$

$$\Rightarrow \int_{-\pi}^{\pi} f(x) dx = \frac{a_0}{2} [x]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} a_n \cdot 0 + \sum_{n=1}^{\infty} b_n \cdot 0$$

$$= \frac{a_0}{2} \cdot 2\pi$$

$$= \pi a_0$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad \text{(Ans)}$$

(ii) multiplying both side $\cos nx$ and limit $\rightarrow \pi$

$$\int_{-\pi}^{\pi} f(x) \cos nx = \frac{a_0}{2} \int_{-\pi}^{\pi} \cos nx dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} a_n \cos nx \cos nx dx + \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} b_n \cos nx \sin nx dx$$

$$\Rightarrow \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{a_0}{2} \cdot 0 + \sum_{n=1}^{\infty} a_n \cdot \pi + 0$$

in each other are 0
other are 20
0 0 0 0 0

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

(iii)

Even/Odd Function

If $f(-x) = f(x)$ Then $f(x)$ is called an even function.

$$f(x) = \cos x$$

$$f(-x) = \cos(-x)$$

$\therefore \cos x$ is even function.

$$\cos x = f(x)$$

odd:

$$f(-x) = -f(x)$$

Then $f(x)$ is called

an odd function



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Sat / Sun / Mon / Tue / Wed / Thu / Fri
Date :

$$f(x) = \sin nx$$

$$f(x) = -\sin(-x) \\ = -f(-x)$$

$\therefore \sin x$ is odd function

* $f(x)$ is even function show that

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_0 = \underbrace{\left[\frac{1}{\pi} \int_{-\pi}^0 f(x) dx \right]}_1 + \underbrace{\left[\frac{1}{\pi} \int_0^{\pi} f(x) dx \right]}_2$$

For (1) =

$$\frac{1}{\pi} \int_{-\pi}^0 f(x) dx$$

$$= -\frac{1}{\pi} \int_{\pi}^0 f(-y) dy$$

$$= \frac{1}{\pi} \int_0^{\pi} f(-y) dy$$

$$= \frac{1}{\pi} \int_0^{\pi} f(-x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} f(x) dx \quad \because f(x) \text{ is even}$$

let
 $x = -y$
 $dx = -dy$

x	$-\pi$	0
y	π	0



Plan Title :

Date :/...../.....

② अब नया same पाओ

$$\frac{1}{\pi} \int_0^{\pi} f(x) dx + \frac{1}{\pi} \int_{\pi}^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

Answer:

$$\int_0^{\pi} \frac{1}{\pi} dx = \frac{1}{\pi} \times \pi = 1$$

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$

$$\left[\int_0^{\pi} \frac{1}{\pi} dx \right] + \left[\int_{\pi}^{\pi} \frac{1}{\pi} dx \right] = 1 + 0 = 1$$

for
x
is

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$

$$\int_{\pi}^{\pi} \frac{1}{\pi} dx = 0$$



Plan Title : Lecture - 08

Sat / Sun / Mon / Tue / Wed / Thu / Fri
☐ ☐ ☐ ☐ ☐ ☐ ☐
 Date : 08/12/2024

Math

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08-12-2024

periodic function if $f(x+\tau) = f(x)$ then $f(x)$ called periodic function and τ is called periodic.

At the end points $x = \pm\pi$ if $f(x)$ is discontinuous then

$$f(\pi) = \lim_{\delta \rightarrow 0} \frac{1}{2} [f(-\pi+\delta) + f(\pi-\delta)]$$

$$f(\pi) = \frac{1}{2} [f(-\pi+0) + f(\pi-0)]$$

(important 100%)

problem solve:

* if $f(x) = x + x^r$ in $(-\pi, \pi)$, Then expand in sine and cosine of the function. and show that

$$\frac{\pi^r}{6} = 1 + \frac{1}{2^r} + \frac{1}{3^r} + \frac{1}{4^r} + \dots$$

Answer:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x + x^r dx$$

$$= \frac{1}{\pi} \left[\frac{x^r}{r} + \frac{x^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^r}{r} + \frac{\pi^3}{3} - \left(\frac{(-\pi)^r}{r} - \frac{\pi^3}{3} \right) \right]$$

$$= \frac{1}{\pi} \cdot \frac{2\pi^3}{3}$$

$$= \frac{2\pi^2}{3}$$

Answer



Plan Title :

Sat / Sun / Mon / Tue / Wed / Thu / Fri
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Date :/...../.....

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \cos nx \, dx$$

$[uv \, dx = u \int v \, dx - \int (u' \int v \, dx) \, dx]$
 २ बाँद apply হবে....

$$= \frac{1}{\pi} \left[(x+x^2) \frac{\sin nx}{n} - \left(1+2x\right) \left(\frac{-\cos nx}{n^2}\right) + 2 \frac{-\sin nx}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{1}{n} (x+x^2) \sin nx + \frac{1}{n^2} (1+2x) \cos nx - \frac{2}{n^3} \sin nx \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left\{ 0 + \frac{1}{n^2} (1+2\pi) (-1)^n - 0 \right\} - \left\{ 0 + \frac{1}{n^2} (1-2\pi) (-1)^n - 0 \right\}$$

$$= \frac{1}{\pi} \frac{(-1)^n}{n^2} (1+2\pi - 1+2\pi)$$

$$= \frac{4\pi (-1)^n}{\pi n^2}$$

$$= \frac{4 (-1)^n}{n^2}$$

$\sin n\pi = 0$
 $\cos n\pi = (-1)^n$

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Plan Title :

$$\int \sin x = -\cos x$$

$$\int \cos x = \sin x$$

Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date :/...../..... (NA)

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \sin nx \, dx$$

$$= \frac{1}{\pi} \left[(x+x^2) \left(\frac{-\cos nx}{n} \right) - (1+2x) \left(\frac{-\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left\{ -\frac{1}{n} (\pi+\pi^2) (-1)^n + 0 + \frac{2}{n^3} (-1)^n \right\} - \left\{ -\frac{1}{n} (-\pi+\pi^2) (-1)^n + 0 + \frac{2}{n^3} (-1)^n \right\} \right]$$

$$= -\frac{1}{n\pi} 2\pi (-1)^n = \frac{-2(-1)^n}{n}$$

Now result,

$$f(x) = \frac{x^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos nx = \sum_{n=1}^{\infty} \frac{2(-1)^n}{n} \sin nx$$

$$= \frac{\pi^2}{3} + 4 \left(-\frac{\cos x}{1} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \dots \right)$$

$$- 2 \left(-\frac{\sin x}{1} + \frac{\sin 2x}{2} - \frac{\sin 3x}{3} + \dots \right)$$

Now

$$f(\pi) = \frac{1}{2} \left\{ f(-\pi+0) + f(\pi-0) \right\}$$

$$= \frac{1}{2} \left\{ -\pi + \pi^2 + \pi + \pi^2 \right\}$$

$$= \pi^2$$

যেহেতু $f(x)=x+x^2$

মানটা এর করে নিলো



Plan Title :

Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date :

$$f(\pi) = \frac{\pi^2}{3} + 4 \left(\frac{-1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \dots \right)$$

+ 2(0)

$f(x) =$
 $x = \text{pai hole}$
 $f(\text{pai})$

$f(\pi) = \pi^2$

$$\pi^2 = \frac{\pi^2}{3} + 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\pi^2 - \frac{\pi^2}{3} = 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\frac{2\pi^2}{3} = 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\frac{\pi^2}{6} = \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

Answer

(3rd page)

$$f(x) = \begin{cases} -\pi & \text{when } -\pi < x < 0 \\ x & \text{when } 0 < x < \pi \end{cases}$$

then

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

① Moved sin

②

Subject: 9/9/25

Date: _____

Fourier Integral :

The condition are called Dirichlet's conditions
Let
① $f(x)$ function satisfy the following conditions
② $f(x)$ is defined in the interval $-l < x < l$
③ $f(x+2l) = f(x)$; i.e. $f(x)$ is periodic and its period is $2l$.

if a function

State Fourier Integral theorem : $\rightarrow$

if a function $f(x)$ satisfied Dirichlet's condition
in the ~~interval~~ interval $(-l, l)$ and $\int_{-\infty}^{\infty} f(x) dx$ is
absolutely convergent. then at each point condition
continuity of $f(x)$ -

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos u(x-t) du dt$$
$$= \frac{1}{\pi} \int_0^{\infty} du \int_{-\infty}^{\infty} f(t) \cos u(x-t) dt$$

if $f(x)$ is even;

$$f(x) = \frac{2}{\pi} \int_0^{\infty} f(t) dt \int_0^{\infty} \cos ut \cos ux du$$

statement and exam 7

Subject: _____

Date: _____

if $f(x)$ is odd then

$$f(x) = \frac{2}{\pi} \int_0^{\infty} f(t) dt$$

★ ★ Fourier transform

★ The function Fourier sine transform:

Let, a function $F(x)$ be defined in the interval $0 < x < l$ then the finite Fourier sine transform of $F(x)$ is denoted by

$F_s[F(x)]$ or $f_s(n)$ and is defined

as follows

$$F_s[F(x)] = f_s(n) = \int_0^l F(x) \sin \frac{n\pi x}{l} dx$$

where n is integer

Reduction
Fourier Transform

Question:

अच्छा reduction गूना
+ Relation Et

Math

Mashud Sir

12-01-25

The finite fourier cosine transform;

Let a Function $F(x)$ be defined in the interval $0 < x < L$. Then the finite fourier cosine transform of $F(x)$ is denoted by $F_c\{F(x)\}$ or $f_c(n)$ and is defined as follows-

$$F_c\{F(x)\} = f_c(n) = \int_0^L F(x) \cos \frac{n\pi x}{L} dx \quad \text{where } n \text{ is an integer.}$$

Inverse finite fourier sine transform;

$$\text{If } f_s(n) = \int_0^L F(x) \sin \frac{n\pi x}{L} dx$$

$$\text{then } F(x) = \frac{2}{L} \sum_{n=1}^{\infty} f_s(n) \sin \frac{n\pi x}{L}$$

is called inverse finite sine transform

Inverse finite fourier cosine transform

$$\text{If } f_c(n) = \int_0^L F(x) \cos \frac{n\pi x}{L} dx$$

Then

$$F(x) = \frac{1}{L} \int_0^L F(x) dx + \frac{2}{L} \sum_{n=1}^{\infty} f_c(n) \cos \frac{n\pi x}{L}$$

$$f_c(n) = \int_0^L F(x) dx$$

(বাক্য ৭)

Fourier Transformer:

$$F\{F(x)\} = f(x) = \int_{-\infty}^{\infty} F(x) e^{inx} dx$$

Inverse Fourier transform:

If $f(x) = \int_{-\infty}^{\infty} F(x) e^{inx} dx$ then

$$F(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-inx} f(x) dx$$

Fourier Sine Transform :

$$F_s\{F(x)\} = f_s(x) = \int_0^{\infty} F(x) \sin x dx$$

must be $x \in \mathbb{R}$ and $x \geq 0$

v.v.i

Relation between Fourier and Laplace transform

Answer:

We consider a Function:

$$F(t) = \begin{cases} e^{-xt} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$F\{F(t)\} = \int_{-\infty}^{\infty} e^{int} F(t) dt$$

$$= \int_{-\infty}^0 e^{int} \cdot 0 dt + \int_0^{\infty} e^{int} F(t) dt$$

$$= \int_{-\infty}^0 e^{int} \cdot 0 dt + \int_0^{\infty} e^{int} e^{-xt} g(t) dt$$



Plan Title :

Sat / Sun / Mon / Tue / Wed / Thu / Fri
□ □ □ □ □ □ □
Date :/...../.....

$$= \int_0^{\infty} e^{(in-x)t} g(t) dt$$

$$= \int_0^{\infty} e^{-(x-in)t} g(t) dt$$

$$\text{let } x-in = s$$
$$= \int_0^{\infty} e^{-st} g(t) dt$$

$$= \mathcal{L}[g(t)]$$

[proved]

important statement
convolution theorem

Q1

Plan Title : Lecture - 10

Sat / Sun / Mon / Tue / Wed / Thu / Fri
☐ ☐ ☐ ☐ ☐ ☐ ☐
 Date :/...../.....

Math

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16 - 01 - 25

v.v.1

EXAMPLE :

$$f(x) = \begin{cases} x & 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

Find the fourier sine transform

Soln :

$$\begin{aligned} F_s \{ f(x) \} &= f_s(n) = \int_0^{\pi} f(x) \sin nx \, dx \\ &= \int_0^{\pi/2} f(x) \sin nx \, dx + \int_{\pi/2}^{\pi} f(x) \sin nx \, dx \\ &= \int_0^{\pi/2} x \sin nx \, dx + \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx \\ &= \left[x \left(\frac{-\cos nx}{n} \right) - \frac{(-\sin nx)}{n^2} \right]_0^{\pi/2} \\ &\quad + \left[(\pi - x) \left(\frac{-\cos nx}{n} \right) - (-1) \frac{-\sin nx}{n^2} \right]_{\pi/2}^{\pi} \\ &= \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi/2} + \end{aligned}$$



Plan Title :

Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date : / /

Exam question : Fourier sine + cos transform question at last

$$\left| \frac{-(\pi-x) \cos nx}{n} - \frac{\sin nx}{n^2} \right|_{\pi/2}^{\pi}$$

$$= -\frac{1}{n} \left(\frac{\pi}{2} \cos \frac{n\pi}{2} - 0 \right) + \frac{1}{n^2} \left(\sin \frac{n\pi}{2} - 0 \right) \\ - \frac{1}{n} \left(0 - \frac{\pi}{2} \cos \frac{n\pi}{2} \right) - \frac{1}{n^2} \left(0 - \sin \frac{n\pi}{2} \right)$$

$$= \frac{2}{n^2} \sin \frac{n\pi}{2} \quad (\text{Ans.})$$

practice :

① If $f(x) = \begin{cases} 1 & 0 < x < \frac{\pi}{2} \\ -1 & \frac{\pi}{2} < x < \pi \end{cases}$ (open interval)

Find the Fourier sine transform and Fourier cos transform

② If $f(x) = e^{-x} \quad x \geq 0$ * Find the Fourier transform

③ If $f(x) = \begin{cases} x^2, & |x| < a \\ 0, & |x| > a \end{cases}$

Find the Fourier transform



Plan Title :

10 marks

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Math

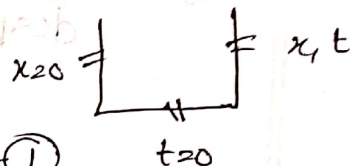
Mashud Sir

2-02-2025

If $u(0, t) = 1$, $u(\pi, t) = 3$, $u(x, 0) = 1$, $u = u(x, t)$

Find the solution of the boundary value problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$



solⁿ:

Given that, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ — (1)

Taking finite Fourier sine transform on both sides of (1), then

we get i.e. $F_s \left\{ \frac{\partial u}{\partial t} \right\} = F_s \left\{ \frac{\partial^2 u}{\partial x^2} \right\}$

$$\int_0^\pi \frac{\partial u}{\partial t} \sin nx \, dx = \int_0^\pi \frac{\partial^2 u}{\partial x^2} \sin nx \, dx \quad \text{--- (5)}$$

consider,

$$u = u(x, t) = \int_0^\pi u(x, t) \sin nx \, dx$$

$$\frac{\partial u}{\partial t} = \int_0^\pi \frac{\partial u}{\partial t} \sin nx \, dx$$

condition

$$u(0, t) = 1 \quad \text{--- (2)}$$

$$u(\pi, t) = 3 \quad \text{--- (3)}$$

$$u(x, 0) = 1 \quad \text{--- (4)}$$

$$= \int_0^\pi \frac{\partial u}{\partial t} \sin nx \, dx$$

[using 5]

$$\frac{\partial u}{\partial t} = \left[\sin nx \frac{\partial u}{\partial x} \right]_0^\pi - n \int_0^\pi \cos nx \frac{\partial u}{\partial x} \, dx$$

$$= 0 - 0 - n \left\{ \cos nx \, u(x, t) \right\}_0^\pi - n \int_0^\pi (-\sin nx) u(x, t) \, dx$$



Plan Title :

$$\frac{dy}{dx} + py = Q$$

Sat / Sun / Mon / Tue / Wed / Thu / Fri

Date :/...../.....

$$= -n \left\{ \cos n\pi \cdot U(\pi, t) - 1 \cdot U(0, t) \right\} - n^2 \int_0^\pi \sin nx \cdot U(x, t) dx$$

$$= -n \left\{ 3 \cos n\pi - 1 \right\} - n^2 U \quad [\text{using 6}]$$

$$\frac{du}{dt} = n \cdot (1 - 3 \cos n\pi) - n^2 u$$

$$\Rightarrow \frac{du}{dt} + n^2 u = n (1 - 3 \cos n\pi) \quad \text{--- (2)}$$

constant

$$\begin{aligned} \int p dx \\ IF = e \\ = e^{\int n^2 dt} \\ = e^{n^2 t} \end{aligned}$$

$$U e^{n^2 t} = \int n (1 - 3 \cos n\pi) e^{n^2 t} dt$$

$$= \frac{n}{n^2} (1 - 3 \cos n\pi) e^{n^2 t} + A$$

$$U = \frac{1 - 3 \cos n\pi}{n} + A e^{-n^2 t}$$

$$U(n, t) = \frac{1 - 3 \cos n\pi}{n} + A e^{-n^2 t} \quad \text{--- (8)}$$

$$U(n, 0) = \frac{1 - 3 \cos n\pi}{n} + A \quad \text{--- (9)}$$

$$U(n, 0) = \int_0^\pi U(x, 0) \sin nx dx$$

$$U(n, 0) = \int_0^\pi 1 \sin nx dx$$

$$f(x) = - \left| \frac{\cos n\pi}{n} \right|_0^{\pi}$$

$$= - \frac{1}{n} (\cos n\pi - 1)$$

$$= (1 - \cos n\pi)$$

from ⑨

$$\frac{1}{n} (1 - 3 \cos n\pi) + A = \frac{1}{n} (1 - \cos n\pi)$$

$$A = \frac{1}{n} (1 - \cos n\pi - 1 + 3 \cos n\pi)$$

$$A = \frac{2 \cos n\pi}{n}$$

$$\textcircled{8} \Rightarrow u(n, t) = \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t}$$

Taking inverse finite fourier sine transformer of with sides then we get

$$u(n, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t} \right\} \sin n\pi$$

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$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t} \right\} \sin n\pi$$

$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t} \right\} \sin n\pi$$

$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t} \right\} \sin n\pi$$

$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{1}{n} (1 - 3 \cos n\pi) + \frac{2 \cos n\pi}{n} e^{-n^2 t} \right\} \sin n\pi$$